FA5_Group1_Quijano_DSC1105

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```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.4.3

## Warning: package 'ggplot2' was built under R version 4.4.3

## Warning: package 'tibble' was built under R version 4.4.3

## Warning: package 'tidyr' was built under R version 4.4.3

## Warning: package 'readr' was built under R version 4.4.3

## Warning: package 'purrr' was built under R version 4.4.3

## Warning: package 'dplyr' was built under R version 4.4.3

## Warning: package 'forcats' was built under R version 4.4.3

## Warning: package 'lubridate' was built under R version 4.4.3

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.1     v stringr   1.5.1
## v ggplot2   4.0.1     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.4
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
library(reshape2)
```

```
## Warning: package 'reshape2' was built under R version 4.4.3
```

```
##
## Attaching package: 'reshape2'
##
## The following object is masked from 'package:tidyr':
##
##     smiths
```

```r
data <- read.csv("adult.csv")
head(data)
```

```
##   age workclass fnlwgt    education education.num marital.status
## 1  90         ?  77053      HS-grad             9        Widowed
## 2  82   Private 132870      HS-grad             9        Widowed
## 3  66         ? 186061 Some-college            10        Widowed
## 4  54   Private 140359      7th-8th             4       Divorced
## 5  41   Private 264663 Some-college            10      Separated
## 6  34   Private 216864      HS-grad             9       Divorced
##          occupation  relationship  race    sex capital.gain capital.loss
## 1                 ? Not-in-family White Female            0         4356
## 2   Exec-managerial Not-in-family White Female            0         4356
## 3                 ?     Unmarried Black Female            0         4356
## 4 Machine-op-inspct     Unmarried White Female            0         3900
## 5    Prof-specialty     Own-child White Female            0         3900
## 6     Other-service     Unmarried White Female            0         3770
##   hours.per.week native.country income
## 1             40  United-States  <=50K
## 2             18  United-States  <=50K
## 3             40  United-States  <=50K
## 4             40  United-States  <=50K
## 5             40  United-States  <=50K
## 6             45  United-States  <=50K
```

```r
data <- data %>%
  mutate(education_group = case_when(
    education %in% c("Preschool", "1st-4th", "5th-6th") ~ "Elementary",
    education %in% c("7th-8th", "9th") ~ "Middle School",
    education %in% c("10th", "11th", "12th", "HS-grad") ~ "High School",
    education %in% c("Some-college", "Assoc-acdm", "Assoc-voc") ~ "College / Associate",
    education %in% c("Bachelors", "Masters", "Doctorate", "Prof-school") ~ "Advanced",
    TRUE ~ NA_character_
  ))

anyNA(data)
```

```
## [1] FALSE
```

```r
table_ed_inc <- table(data$education_group, data$income)

table_with_totals <- addmargins(table_ed_inc)

row_percent <- prop.table(table_ed_inc, margin = 1) * 100

combined <- cbind(table_ed_inc, round(row_percent, 2))
```

```r
apa_table <- as.data.frame.matrix(table_ed_inc) %>%
  mutate(Total = rowSums(.)) %>%
  mutate(`<=50K (%)` = paste0(`<=50K`, " (", round(`<=50K`/Total*100, 2), "%)"),
         `>50K (%)` = paste0(`>50K`, " (", round(`>50K`/Total*100, 2), "%)")) %>%
  select(`<=50K (%)`, `>50K (%)`, Total)

apa_table
```

```
##                          <=50K (%)      >50K (%) Total
## Advanced              4158 (51.54%) 3909 (48.46%)  8067
## College / Associate   7727 (79.33%) 2013 (20.67%)  9740
## Elementary             530 (96.01%)    22 (3.99%)   552
## High School          11212 (85.97%) 1830 (14.03%) 13042
## Middle School         1093 (94.22%)    67 (5.78%)  1160
```

```r
chisq_test = chisq.test(table(data$education_group, data$income))

chi_stat <- chisq_test$statistic
chi_dof <- chisq_test$parameter
chi_p <- chisq_test$p.value

print("A chi-square test of independence showed a significant association between education level and i
```

```
## [1] "A chi-square test of independence showed a significant association between education level and i
```

```r
tab_ed_inc <- table(data$education_group, data$income)
tab_ed_inc
```

```
##
##                        <=50K  >50K
##   Advanced              4158  3909
##   College / Associate   7727  2013
##   Elementary             530    22
##   High School          11212  1830
##   Middle School         1093    67
```

```r
chisq_test <- chisq.test(tab_ed_inc)

expected <- chisq_test$expected

std_res <- chisq_test$stdres

round(expected, 2)
```

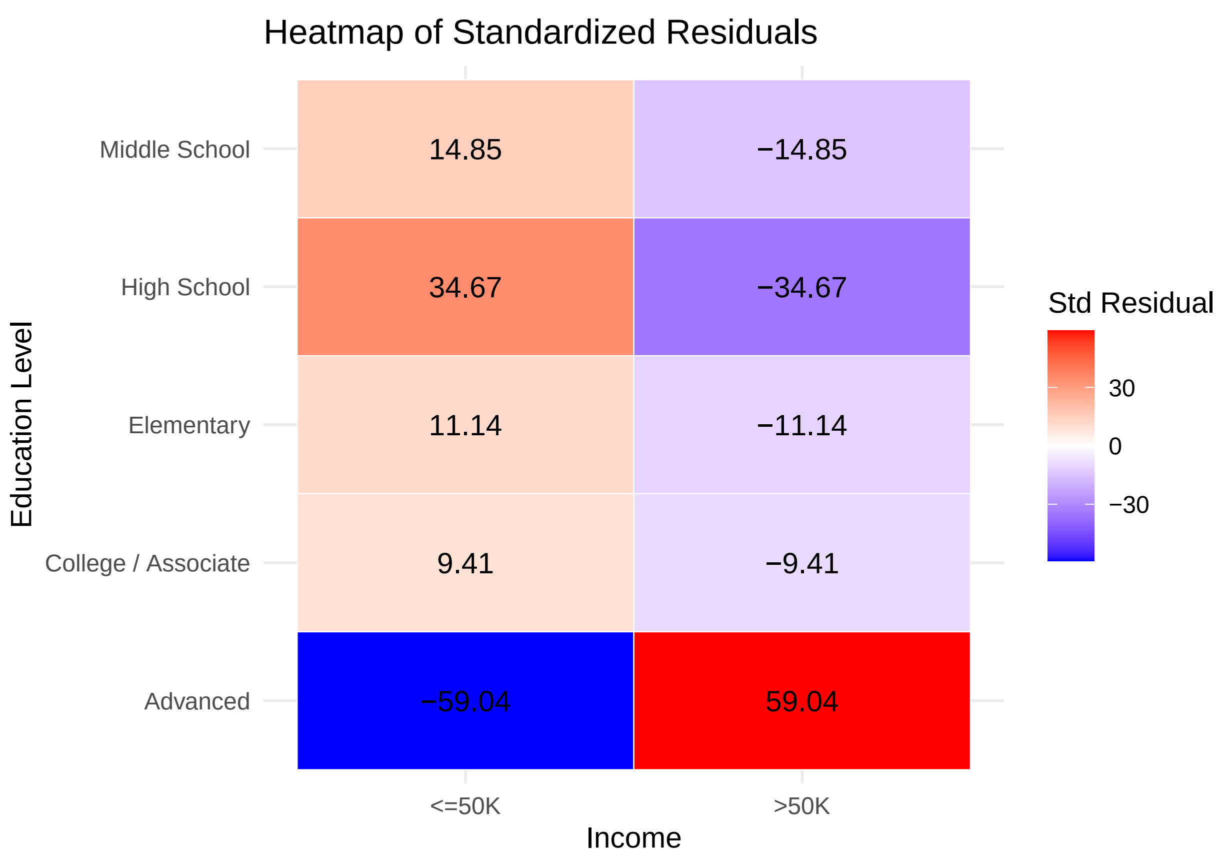
```
##
##                          <=50K    >50K
##   Advanced             6124.39 1942.61
##   College / Associate  7394.51 2345.49
##   Elementary            419.07  132.93
##   High School          9901.36 3140.64
##   Middle School         880.66  279.34
```

```r
round(std_res, 2)
```

```
##
##                       <=50K   >50K
##   Advanced           -59.04  59.04
##   College / Associate  9.41  -9.41
##   Elementary          11.14 -11.14
##   High School         34.67 -34.67
##   Middle School       14.85 -14.85
```

```r
res_df <- melt(std_res)

ggplot(res_df, aes(x = Var2, y = Var1, fill = value)) +
  geom_tile(color = "white") +
  geom_text(aes(label = round(value, 2)), size = 4) +
  scale_fill_gradient2(low = "blue", mid = "white", high = "red", midpoint = 0) +
  labs(
    title = "Heatmap of Standardized Residuals",
    x = "Income",
    y = "Education Level",
    fill = "Std Residual"
  ) +
  theme_minimal()
```



a. A chi-square test of independence revealed a significant association between education level and income, $X\hat{}2(df = 4) = 3738.87$, $p < 0.05$. This suggests that income distribution varies across education levels.

b. All categories show significant deviations, but key ones:

Advanced (>50K): $z = 59.04$ → far more high-income individuals than expected

Advanced (<=50K): $z = -59.04$ → far fewer low-income individuals than expected

High School (<=50K): $z = 34.67$ → overrepresented in lower income

High School (>50K): $z = -34.67$ → underrepresented in higher income

Elementary & Middle School also strongly:

Overrepresented in <=50K

Underrepresented in >50K

c. Higher education strongly predicts higher income — individuals in the Advanced category are dramatically overrepresented in the >50K group.

Lower education levels are concentrated in lower income brackets, especially Elementary and Middle School groups.

High School and College/Associate groups fall in between, but still skew toward <= 50K income.